

The IAD Seminar Series

Innovation Across Disciplines

Title: Damage detection in structures by stress waves- a time series-based approach

Abstract: This presentation explores the application of time series model-based system identification for detecting structural damage through stress wave propagation. By integrating statistical time series analysis with physics-based wave propagation principles, the approach enables robust detection of subtle structural changes and material degradation. The presentation further examines the use of stress waves as a non-destructive sensing mechanism for identifying aging and state-dependent variations in lithium-ion batteries, with implications for real-time battery health monitoring and prognostics. Finally, the presentation discusses fundamental challenges in accurately simulating stress wave propagation in complex structural systems, including multiphysics interactions, material heterogeneity, and computational limitations.

Zoom: <https://sdstate.zoom.us/j/92066880758>



**SOUTH DAKOTA
STATE UNIVERSITY**

Time and location: 12-1 PM Thursday, 2/26/2026, Crothers 307
Pizza is provided!!!

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